

FACULTY PROSPECTUS · RESEARCH PLATFORM PROPOSAL

LA Water Security Atlas

Understanding the Spatial Vulnerability of
Southern California's Water System

Kai Nozawa · Independent public geography project · Santa Monica College GIS craft applied as method

Prepared for informal faculty review · Not a course assignment

Not a news site — a research platform anyone can read

A research platform that makes Los Angeles's water system understandable — and lets users see spatial vulnerability for themselves.

Design principle

Show data so users understand risk themselves — not a lecture that tells them what to fear.

Entry question

Where does Los Angeles get its water?
Click → metropolitan Los Angeles map (not a world map).

Academic fit

Human–environment interaction · water security · infrastructure · climate risk · scenario analysis.

Why Los Angeles water

Los Angeles is a metropolitan water system produced across basins — not a city that drinks only local rain.

01

Multi-source dependency (LADWP)

Primary supplies: Los Angeles Aqueduct (Eastern Sierra), MWD purchases (SWP + Colorado River), local groundwater, and expanding recycled / stormwater capture (LADWP UWMP 2020; Briefing Book 2022–23).

02

L.A. Aqueduct is hydrologically volatile

LAA share of city supply has ranged roughly 10–50% with snowpack; recent years ~230,000 AF/yr (~75 billion gal). About half of historically exported Eastern Sierra water is now left for Owens/Mono environmental obligations (LADWP).

03

Dry years raise import dependence

When Sierra snowpack falls, LAA deliveries drop and MWD purchases rise — an inverse relationship that maps vulnerability onto distant SWP and Colorado allocations (LADWP Securing L.A.'s Water Supply, 2008; UWMP series).

04

Interbasin transfer as urban geography

Metropolitan growth depended on moving water across mountains and deserts (Hundley 2001; Kahrl 1982). The tap is local; the production of water is regional and multi-state.

Why spatial vulnerability

Vulnerability is not only “less water.” It is where risk is produced, how it travels along corridors, and who inherits it.

01

Climate risk is spatially transmitted

Southwest water security is tightly coupled to warming, snowpack decline, and reservoir stress — risks that enter Southern California through shared rivers and mountain source regions (MacDonald 2010; Gleick 2010).

02

Colorado River over-allocation

The 1922 Compact and later operations allocate a river whose twentieth-century assumptions overstate reliable flow; Mead/Powell shortage is a basin-scale stress that travels via the CRA into Southern California (Barnett & Pierce 2008; USBR shortage operations).

03

Infrastructure & energy exposure

Long aqueducts require lift and pumping (e.g., Edmonston / Tehachapi on the SWP). Supply reliability is coupled to energy systems — outage and cost become water-security variables, not side notes.

04

Public literacy gap = research opportunity

Agency plans document mix and constraints; public tools rarely let users see cascading spatial risk or compare scenarios without false precision. That is the Atlas design gap.

Key sources: LADWP UWMP / Briefing Book · Hundley 2001 · Kahrl 1982 · MacDonald 2010 · Barnett & Pierce 2008 · Gleick 2010 · Colorado River Compact / USBR

From Where to Why — and What if

Primary RQ: How is Southern California's urban water supply spatially produced and institutionally organized, and what vulnerabilities become visible when climate, drought, population, and energy stress are mapped together?

RQ1 · Where

Which sources and corridors supply Los Angeles, and how do their geographies differ?

RQ2 · Why

Why does dependency persist — history, policy, agriculture, interstate compact, urban growth?

RQ3 · Risk

How do snowpack, drought, population, and energy constraints reshape vulnerability over time?

RQ4 · Scenarios

How do alternative futures compare under uncertainty — without claiming a single forecast?

Seven layers + simulation — one GIS foundation



+ Core features: Scenario Simulation · Policy Simulator · “Why” panels (history · policy · compact)

Sources · Process · Infrastructure

Layer 1 · Sources

Color-coded supply origins:

- Colorado River
- Owens Valley / Eastern Sierra
- State Water Project
- Local groundwater
- Recycled water

Click → annual volume, share ranges, seasonal pattern, recent trend — with uncertainty notes.

Layer 2 · How water is made

Correct the common myth: L.A. water is not “created.”
It is collected, treated, and conveyed.

Snowpack → Reservoir → Aqueduct → Treatment →
Distribution → House

Explorable 3D process view for public literacy.

Layer 3 · Infrastructure

Asset dossiers (e.g. Colorado River Aqueduct):

- Length · pumping plants · elevation lift
- Energy use · build year · O&M context
- Photos · video · Street View links

Makes infrastructure a first-class geographic object.

Climate · Drought · Population · Energy

Layer 4 · Climate

Snowpack series (1980–present) beside reservoir relationship — climate signal made spatial.

Layer 5 · Drought

Episode path: 2012 → 2015 → 2022 → 2035 scenarios. Stress as history + futures, not a single year.

Layer 6 · Population

Demand overlay: growth → water demand → shortage pressure. Couples demography to supply geography.

Layer 7 · Energy

Conveyance is energy-intensive. Outage → pumping stop → supply risk. Links water security to energy security.

Scenario analysis — compare futures, do not declare one

Climate, population, and policy are uncertain. Research and policy analysis rely on “what if” comparison — not single-point prediction.

Stress Simulation

Sliders (examples):

- Snowpack 100% → 70% → 40% → 20%
- Colorado River -20%

Map outputs update for reservoirs, deliveries, agriculture, households, and power demand — so users see cascading spatial effects.

Policy Simulator (2035)

Example: population +15%.

Compare portfolios:

- Needed water vs shortfall
- Recycled water +30%
- Desalination option
- Groundwater recovery

Goal: transparent trade-offs under shared assumptions.

GIS product → research platform: answer Why

Berkeley / UCLA Geography evaluates not only map craft, but whether the work explains structure, power, history, and institutions.

History

How Owens, Colorado, and SWP corridors were built into metropolitan life

Policy & law

Compacts, court limits, local agreements, wholesale institutions (MWD, LADWP, DWR)

Economy

Agriculture, urban growth, and demand management as geographic forces

Energy–water nexus

Lift, pumping, and outage risk as coupled infrastructure security

Authoritative sources · transparent methods · open stack

Indicative data sources

- LADWP Urban Water Management Plans
- Metropolitan Water District (MWD)
- California DWR / State Water Project
- U.S. Bureau of Reclamation (Colorado)
- USGS / state hydrography & basins
- NASA / climate & snow products (as appropriate)
- U.S. Census / ACS (population & demand context)

All layers: access dates, limits, and no false-precision shares.

Technical stack (v1 → v2)

v1 (live prototype):

Static map essay + scroll-linked corridors

v2 platform:

- Web GIS (MapLibre / ArcGIS)
- Time-series dashboards
- Scenario engine (rules + models)
- Optional LLM “explain Why” panels grounded in cited sources
- Hosting: static + serverless APIs

Ethics: schematic corridors labeled; not a legal rights atlas.

Phase 1–3 — then California Resource Security Atlas

Phase 1

Public water story + methods

Ship legible source→tap narrative; academic sources; faculty feedback on scope & cartography.

Phase 2

Atlas layers + dossiers

Sources, process, infrastructure panels; climate/drought time series; energy–water link.

Phase 3

Simulation + policy compare

Snowpack/Colorado sliders; 2035 portfolio compare; documented assumptions & uncertainty.

Horizon: California Resource Security Atlas — Water · Energy · Wildfire · Housing · Ports · Earthquakes — same GIS spine.

Why this fits UC Berkeley & UCLA Geography

UC Berkeley Geography

Emphasizes human–environment systems, critical infrastructure, climate adaptation, and spatial analysis that asks why patterns exist.

This project pairs public cartography with scenario analysis and institutional explanation — not decoration maps.

UCLA Geography

Strong tradition in urban environment, water resources of the Southwest, GIS, and applied spatial science in Southern California.

L.A. as study area + regional hydrosocial system is a natural fit for portfolio and later research.

What success looks like

Public literacy

A non-specialist can explain L.A. water sources, corridors, and key risks after using the site.

Research readiness

Transparent methods, sources, and scenario assumptions suitable for faculty critique.

Portfolio signal

Transfer-ready artifact: independent, spatial, policy-aware, expandable beyond water.

Social value

Supports education and civic discussion without claiming official agency endorsement.

What I am asking faculty

1. Is the Phase 1 → Phase 3 scope realistic for a strong independent project?
2. Which datasets are most trustworthy for public synthesis vs. research-grade analysis?
3. Where is the ethical line between helpful simplification and misleading cartography?
4. Which “Why” panels (history, compact, energy–water) should be prioritized first?
5. Would GIS 27 / Independent Study / informal mentorship be appropriate later?